

# **IPL Crunch '26- Data Analysis Report**

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## **1. Introduction**

### **Objective of the Analysis**

The objective of this project is to analyze IPL match data using ball-by-ball statistics to uncover meaningful insights related to match outcomes, team strategies, player performances, and overall competitiveness in the tournament.

The analysis focuses on understanding:

- The impact of winning the toss on match results
- The influence of different innings phases on winning
- Batting first vs batting second advantages
- Team decision-making after winning the toss
- Match competitiveness and close finishes
- Top-performing batters and bowlers across five IPL seasons

The project combines data preprocessing, statistical analysis, and visualization techniques to identify patterns and trends within IPL matches.

## **2.Dataset Used**

The dataset used for this analysis was obtained from Cricsheet in JSON format and contains detailed ball-by-ball IPL match data. Each row in the processed dataset represents a single delivery bowled during a match and includes information such as batting team, bowler, batter, runs scored, wickets, innings, toss results, and match winners.

### **JSON to CSV Conversion**

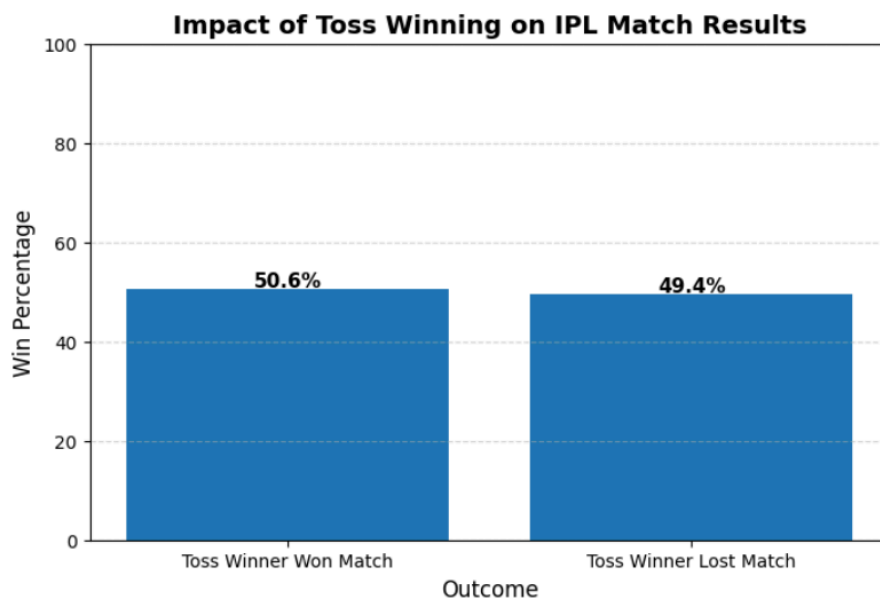
The JSON files were parsed using Python libraries such as `json` and `pandas`. Important match-level and ball-by-ball information including innings, batting team, runs, wickets, toss winner, toss decision, and match winner were extracted and converted into a structured tabular format (CSV) for easier analysis.

### 3.Toss Impact Analysis

Does winning the toss really decide IPL matches?

The analysis shows that teams winning the toss won **50.6%** (Fig 1) of matches, while teams losing the toss still managed to win **49.4%** (Fig 1) of matches. The difference is extremely small, suggesting that toss advantage has minimal impact on overall match results.

**Key Takeaway:** The IPL is far more influenced by team performance, strategy, and execution than by toss luck. Even after losing the toss, teams remain almost equally likely to win the match.



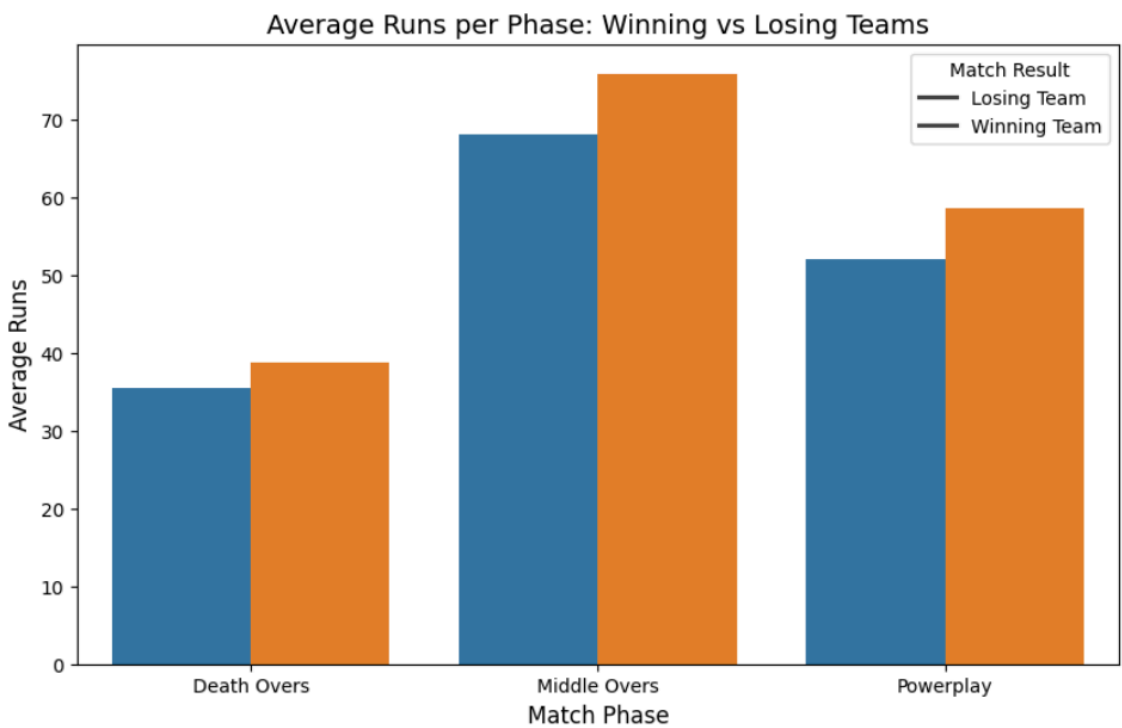
**Figure 1:** Toss-winning impact on IPL match outcomes.

## 4.Phase-wise Performance Analysis

This chart compares the average runs scored by winning and losing teams across different phases of an IPL innings.

Winning teams consistently scored more runs in every phase, with the biggest difference appearing during the **middle overs**. This highlights how maintaining momentum between the powerplay and death overs plays a crucial role in securing victories.

**Key Takeaway:** Contrary to the common belief that death overs decide T20 matches, the analysis suggests that strong middle-over performance has a greater impact on match outcomes.



**Figure 2:** Average runs scored by winning and losing teams across different match phases.

## 5.Top Player Analysis

The table highlights the top-performing batters and bowlers based on total runs scored and wickets taken across the selected IPL seasons.

These players consistently delivered impactful performances and played a major role in shaping match outcomes throughout the tournament.

**Key Takeaway:** The analysis shows that long-term consistency and adaptability are key traits shared by the IPL's most successful players.

[47]:

| Top 5 IPL Batters and Bowlers |      |           |         |
|-------------------------------|------|-----------|---------|
| Batter                        | Runs | Bowler    | Wickets |
| V Kohli                       | 9155 | YS Chahal | 238     |
| RG Sharma                     | 7291 | B Kumar   | 237     |
| S Dhawan                      | 6769 | SP Narine | 225     |
| DA Warner                     | 6567 | DJ Bravo  | 207     |
| KL Rahul                      | 5712 | JJ Bumrah | 207     |

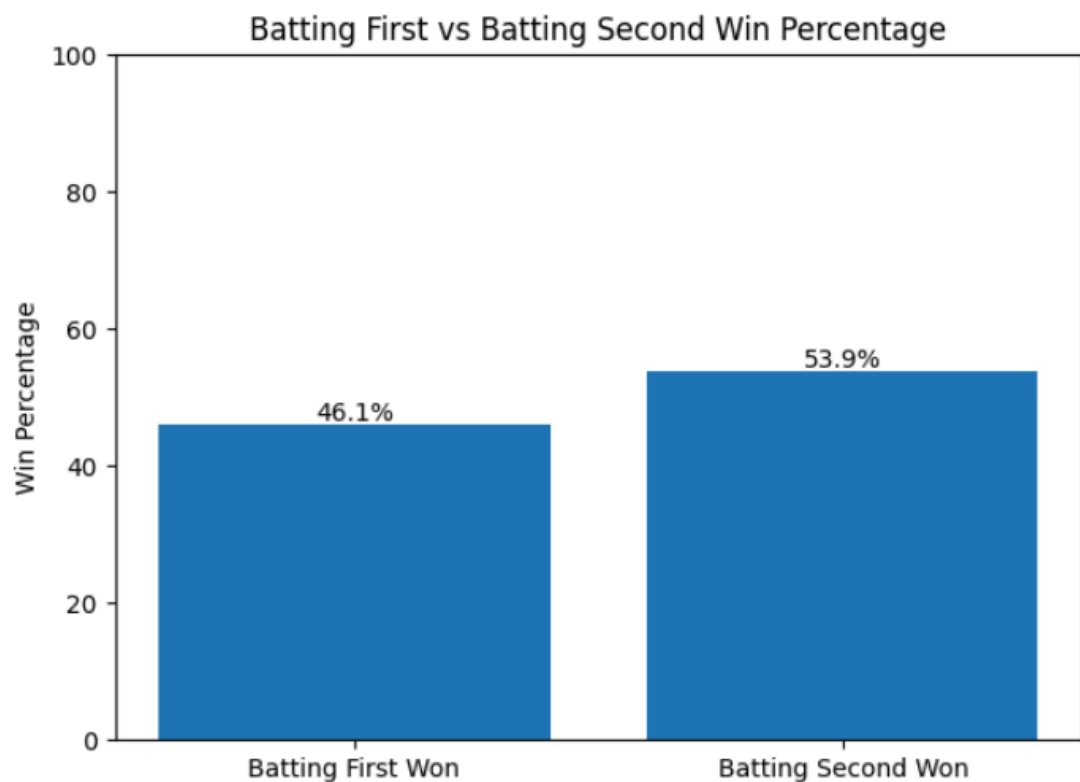
**Table 1:** Top 5 batters by runs and top 5 bowlers by wickets across five IPL seasons.

## 6. Batting First vs Batting Second Analysis

The analysis compares the winning percentages of teams batting first and teams batting second across five IPL seasons.

The results show that teams batting second won **53.9%** of matches, compared to **46.1%** for teams batting first. This indicates a noticeable advantage for chasing teams in IPL matches.

**Key Takeaway:** The data suggests that chasing a target provides teams with a strategic advantage, likely due to better decision-making under known match conditions and scoreboard pressure management.



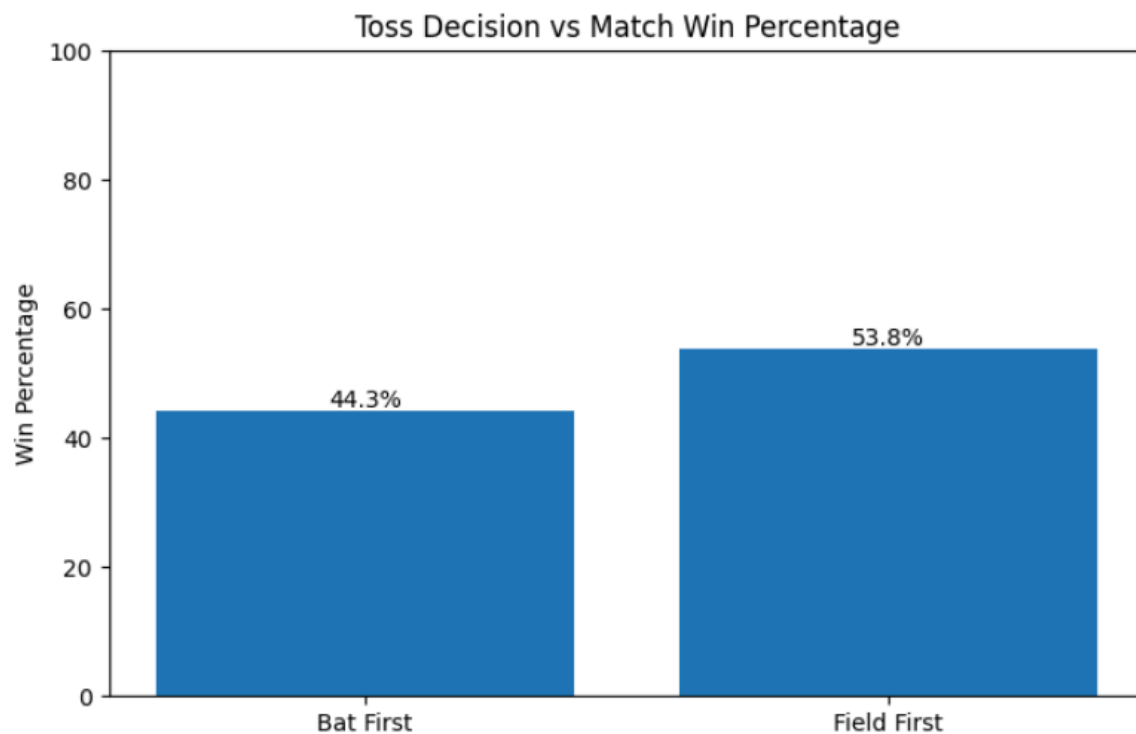
**Figure 3:** Winning percentage comparison between teams batting first and batting second.

## 7. Toss Decision Analysis

This analysis compares the success rates of teams after choosing to bat first or field first upon winning the toss.

The results show that teams choosing to field first won **53.8%** of matches, while teams choosing to bat first won only **44.3%**. This highlights a clear advantage for teams preferring to chase targets in IPL matches.

**Key Takeaway:** The data suggests that captains benefit more from bowling first after winning the toss, likely due to improved chasing strategies and better adaptation to match conditions during the second innings.



**Figure 4:** Match-winning percentage based on toss decisions across five IPL seasons.

## 8.Close Match Analysis

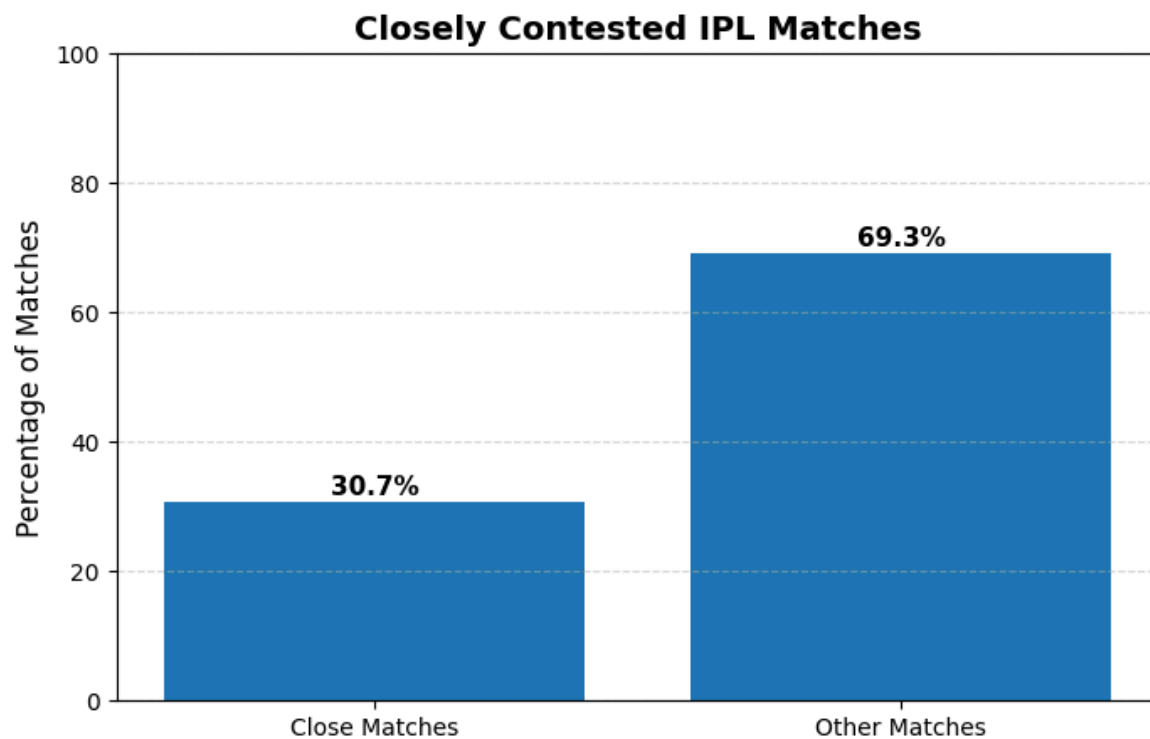
This analysis identified closely contested IPL matches using two conditions: matches won by 10 runs or fewer while defending a target, or successful chases completed during the final over.

The results showed that approximately **30.7%** of IPL matches were closely contested, highlighting how frequently teams remain competitive until the final stages of the game.

### Key Takeaway

One of the most surprising findings was that nearly one-third of IPL matches turned into high-pressure finishes, emphasizing the unpredictability and excitement associated with T20 cricket.

**Figure 5:** Percentage of closely contested IPL matches across five IPL seasons.



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## **9.Best Final One-Line Insight**

**One surprising finding was that nearly one-third of IPL matches turned into close finishes, highlighting the highly competitive and unpredictable nature of the IPL.**

## **Conclusion**

This project analyzed ball-by-ball IPL data from five seasons to identify patterns related to toss impact, batting strategy, match phases, player performance, and match competitiveness.

The analysis revealed that winning the toss alone has minimal influence on match outcomes, while factors such as chasing strategy, middle-over performance, and consistent execution play a much larger role in determining victories. The study also highlighted how frequently IPL matches produce close finishes, demonstrating the highly competitive nature of the tournament.

Overall, the project shows how data analysis can uncover valuable cricket insights and transform raw match data into meaningful strategic observations.

# **Thank You**